

## Ch2. Simple Linear Regression

- Relationship between 2 variables
- The regression model
- Assumptions
- Estimation and method of least squares
- Inferences concerning  $\beta_1$  and  $\beta_0$
- Estimation of the mean of the response variable for a given level of  $x$
- Prediction of new observation
- Analysis of variance approach to regression analysis
- Measures of linear association between  $x$  and  $y$

# Simple Linear Regression Model

*dependent variable (response)*  
*independent variable (explanatory variable, predictor)*  

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, 2, \dots, n$$
*random error*

## Assumptions:

- $E(\epsilon_i) = 0$ ,
- $\text{Var}(\epsilon_i) = \sigma^2$  — *constant variance*
- $\text{Cov}(\epsilon_i, \epsilon_j) = 0$

$$\mathcal{D} = \{(x_i, y_i), i=1, \dots, n\}$$

## ■ In matrix notation.

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

## Simple Linear Regression Equation

*predicted value*

*prediction*

*estimate of  $\beta_0$*

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- The simple linear regression equation provides an estimate of the population regression line
- $\hat{\beta}_0$  is the estimated average value of  $y$  when the value of  $x$  is zero
- $\hat{\beta}_1$  is the estimated change in the average values of  $y$  as a result of a one-unit change in  $x$

## Simple Linear Regression: an example

A real estate agent wishes to examine the relationship between the selling price of a home and its size (measured in square feet)

- A random sample of 10 houses is selected
- $y$  = house price in \$1000s,  $x$  = square feet

| $y$ | $x$  |
|-----|------|
| 245 | 1400 |
| 312 | 1600 |
| 279 | 1700 |
| 308 | 1875 |
| 199 | 1100 |
| 219 | 1550 |
| 405 | 2350 |
| 324 | 2450 |
| 319 | 1425 |
| 255 | 1700 |

$$X = c(1400, 1600, \dots, 1700)$$

$$Y = c(245, \dots, 255)$$

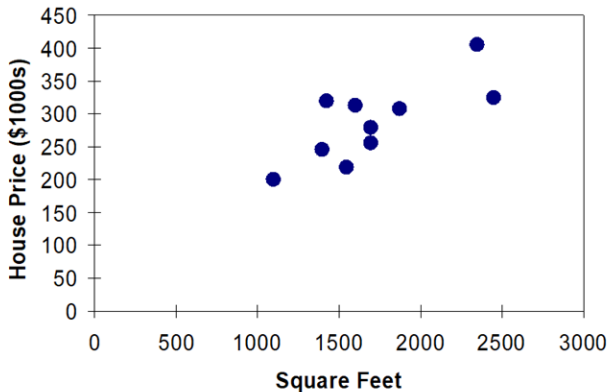




## An example: Graphical Presentation

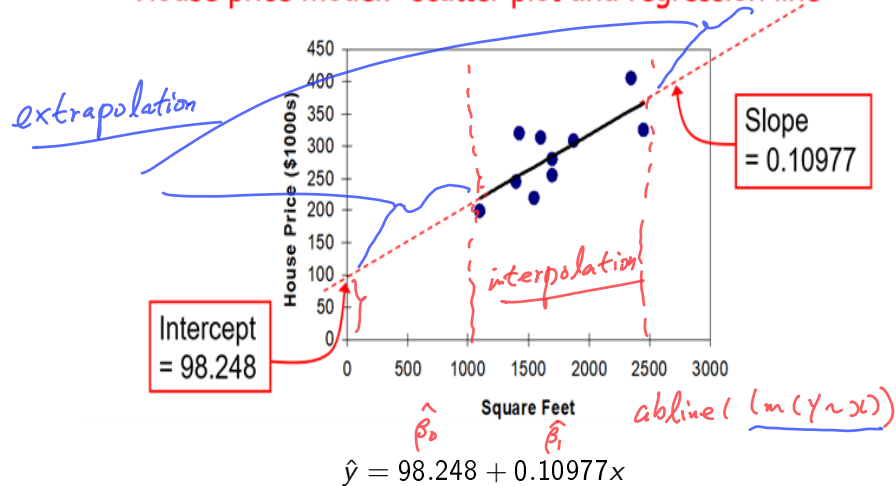
*plot(x, y)*

House price model: scatter plot



## An example: Graphical Presentation

House price model: scatter plot and regression line



## An example: Interpretation of the intercept, $\hat{\beta}_0$

$$\hat{y} = 98.248 + 0.10977x$$

- $\hat{\beta}_0$  is the estimated average value of  $y$  when the value of  $x$  is zero (if  $x = 0$  is in the range of observed  $x$  values)
- Here, no houses had 0 square feet, so  $\hat{\beta}_0 = 98.248$  just indicates that, for houses within the range of sizes observed, \$98,248 is the portion of the house price not explained by square feet.

## An example: Interpretation of the Slope Coefficient, $\hat{\beta}_1$

$$\hat{y} = 98.248 + 0.10977x$$

- $\hat{\beta}_1$  measures the estimated change in the average value of  $y$  as a result of a one-unit change in  $x$ 
  - Here,  $\hat{\beta}_1 = .10977$  tells us that the average value of a house increases by  $.10977(k) = \$109.77$ , on average, for each additional one square foot of size.

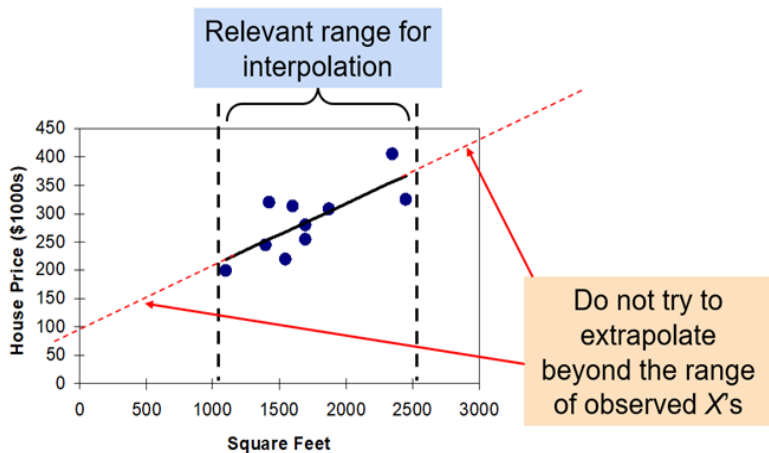
## An example: Predictions using Regression Analysis

- Predict the price for a house with 2000 square feet:

$$\hat{y} = 98.25 + 0.10977 \times 2000 = 317.85$$

- The predicted price for a house with 2000 square feet is  
 $317.85(\$1,000s) = \$317,850$

## An example: Interpolation vs. Extrapolation



When using a regression model for prediction, only predict within the relevant range of data unless you have further information.

## Estimation: Method of Least Squares

- $\hat{\beta}_0$  and  $\hat{\beta}_1$  are obtained by finding the values of  $\beta_0$  and  $\beta_1$  that minimize the sum of the squared differences between  $y$  and  $\hat{y}$ :

$$\min \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Solutions:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

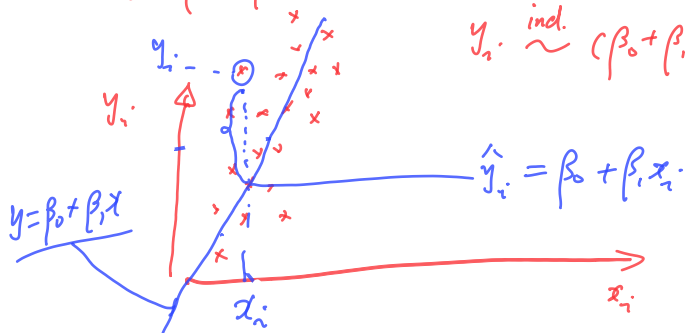
- Comparing  $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$  with  $r = \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}}$

Remark 2.1

Data:  $(x_i, y_i)$ ,  $i=1, \dots, n$ .

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{iid}{\sim} (0, \sigma^2)$$

$$y_i \stackrel{ind.}{\sim} (\beta_0 + \beta_1 x_i, \sigma^2)$$



"1) For the  $i$ -th observation, the difference (distance) between the observed data  $y_i$  (given  $x_i$ ) and the predicted value  $\hat{y}_i$  (given  $x_i$ )

$$(y_i - \hat{y}_i)^2 = (y_i - \beta_0 - \beta_1 x_i)^2 \quad \text{--- } L_2 \text{ distance}$$

$$|y_i - \hat{y}_i| \quad \text{--- } L_1 \text{ distance}$$

"2) LSE: to find  $\hat{\beta}_0$  and  $\hat{\beta}_1$ , s.t.

$$\underline{SSE} = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 = \min$$

(sum of the squared errors)

method of least square

$$\begin{cases} \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \\ \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \end{cases}$$

$$\begin{aligned} (3) \quad 0 = \frac{\partial SSE}{\partial \beta_0} &= -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = -2 \left( \sum y_i - n\beta_0 - \beta_1 \sum x_i \right) \\ &= -2n(\bar{y} - \beta_0 - \beta_1 \bar{x}) \quad (*) \end{aligned}$$

$$0 = \frac{\partial SSE}{\partial \beta_1} = -2 \sum_{i=1}^n [x_i (y_i - \beta_0 - \beta_1 x_i)]$$



$$= -2 \left( \sum_{i=1}^n x_i y_i - \beta_0 \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 \right) \quad (**)$$

$$(*) \Rightarrow \beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$(**) \Rightarrow 0 = \sum_{i=1}^n x_i y_i - \underbrace{(\bar{y} - \beta_1 \bar{x})}_{\beta_0} \underbrace{\sum_{i=1}^n x_i}_{= n\bar{x}} - \beta_1 \sum_{i=1}^n x_i^2$$

$$= \left( \sum_{i=1}^n x_i y_i - n \bar{x} \bar{y} \right) - \beta_1 \left( \sum_{i=1}^n x_i^2 - n \bar{x}^2 \right)$$

$$\Rightarrow \hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}}{\sum_{i=1}^n x_i^2 - n \bar{x}^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{s_{xy}}{s_{xx}}$$

$$\Rightarrow \begin{cases} \hat{\beta}_1 = \frac{s_{xy}}{s_{xx}} \\ \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \end{cases}$$

In theory, we also need to prove the above solutions is the minimum of

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$$

$$\frac{\partial^2 SSE}{\partial \beta^2} \bigg|_{\hat{\beta}} \geq 0$$

How to estimate  $r^2$ ?

Recall:

$$y_1, \dots, y_n \sim N(\mu, \sigma^2)$$

$$s^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{\underbrace{n-1}_{\text{degree of freedom}}}$$

$$s^2 = \hat{\sigma}^2$$

$$E(s^2) = \sigma^2$$

$$y_1, \dots, y_n \stackrel{\text{ind}}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2)$$

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$S^2 = \frac{SSE}{n-2}$$

$$E(S^2) = \sigma^2$$

We can prove:  $\frac{(n-2)S^2}{\sigma^2} \sim \chi^2_{n-2}$

(we will give the proof later)

In Assignment 1, you will need to prove

$\bar{y}$  and  $\hat{\beta}_1$  are independent,

## Estimation of error terms variance $\sigma^2$

- The estimator of  $\sigma^2$  is

$$S^2 = MSE = \frac{SSE}{n-2} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}$$

- $S^2$  is an unbiased estimator of  $\sigma^2$

## Estimation: Method of Maximum Likelihood

- The simple linear regression model with normal error

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2) \quad i = 1, 2, \dots, n,$$

- The likelihood of the above model
- $\hat{\beta}_0$  and  $\hat{\beta}_1$  are obtained by maximising the above likelihood
- MLEs:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

- The estimator of  $\sigma^2$  is  $\frac{SSE}{n} = \frac{n-2}{n} S^2$ .

Remark 2.2      MLE of  $(\beta_0, \beta_1, \sigma^2)$

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

$$\Rightarrow y_i \stackrel{ind.}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2)$$

$$p(y_i) = ?$$

How to find MLE?

— likelihood of  $(y_1, \dots, y_n)$

$$p(y_1, \dots, y_n) = p(y_1) \cdot p(y_2) \cdots p(y_n)$$

— MLE: to find  $(\beta_0, \beta_1, \sigma^2)$  by maximizing likelihood.

$\Leftrightarrow$  maximizing the log-likelihood.

$$l = \log p(y_1, \dots, y_n)$$

$$= \sum_{i=1}^n \log p(y_i)$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2} - \frac{\sum_{i=1}^n (x_i - \bar{x}) \bar{y}}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$= \frac{\sum_{i=1}^n c_i y_i}{\sum_{i=1}^n c_i^2}, \quad c_i = \frac{x_i - \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

N.T.P  $E(\hat{\beta}_1) = \beta_1, \quad \text{Var}(\hat{\beta}_1) = ?$

$$\hat{\beta}_1 \sim N(E(\hat{\beta}_1), \text{Var}(\hat{\beta}_1))$$

$$E(\hat{\beta}_1) = \sum_{i=1}^n c_i (E y_i) = \sum_{i=1}^n c_i (\beta_0 + \beta_1 x_i)$$

$$= \beta_0 \cdot \sum_{i=1}^n c_i + \beta_1 \sum_{i=1}^n c_i x_i$$

$$= \beta_1$$

$$\text{Var}(\hat{\beta}_1) = \sum_{i=1}^n c_i^2 \cdot \text{Var}(y_i)$$

$$= \sigma^2 \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \stackrel{d}{=} \frac{\sigma^2}{S_{xx}}$$

$$\begin{cases} \sum c_i = 0 \\ \sum c_i x_i = \sum_{i=1}^n \frac{(x_i - \bar{x}) x_i}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ = \sum_{i=1}^n \frac{(x_i - \bar{x}) (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ = 1 \end{cases}$$

$$\hat{\beta}_1 \sim N(\beta_1, \sigma^2/s_{xx})$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \sim N(\quad, \quad)$$

$$\begin{aligned} E(\hat{\beta}_0) &= E\bar{y} - E(\hat{\beta}_1) \cdot \bar{x} \\ &= \beta_0 + \beta_1 \bar{x} - \beta_1 \bar{x} \\ &= \beta_0 \end{aligned}$$

$$\begin{aligned} \bar{y} &= \frac{1}{n} \sum y_i \\ E\bar{y} &= \frac{1}{n} \sum E(y_i) \\ &= \frac{1}{n} \sum (\beta_0 + \beta_1 x_i) \\ &= \beta_0 + \beta_1 \bar{x} \end{aligned}$$

$$\begin{aligned} \text{Var}(\hat{\beta}_0) &= \text{Var}(\bar{y}) + \bar{x}^2 \text{Var}(\hat{\beta}_1) - 2\bar{x} \underbrace{\text{cov}(\bar{y}, \hat{\beta}_1)}_{!! \text{ (Q1c, Ans 1)}} \\ &= \frac{\sigma^2}{n} + \bar{x}^2 \frac{\sigma^2}{s_{xx}} = \left( \frac{1}{n} + \frac{\bar{x}^2}{s_{xx}} \right) \sigma^2 \end{aligned}$$

$$\hat{\beta}_0 \sim N\left(\beta_0, \left(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}}\right) \sigma^2\right)$$

$$\frac{\lambda^2}{\hat{V}_{MSE}} = \frac{SSE}{n} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n} = \frac{n-2}{n} s^2 =$$

$$E\left(\frac{\lambda^2}{\hat{V}_{MSE}}\right) = \frac{n-2}{n} \sigma^2 \rightarrow \sigma^2 \quad \text{as } n \rightarrow \infty$$

$\frac{\lambda^2}{\hat{V}_{MSE}}$  is an asymptotic unbiased estimate of  $\sigma^2$ .

## Estimation: Method of Maximum Likelihood

- MLE of  $\beta_0$  = LSE of  $\beta_0$  and is unbiased
- MLE of  $\beta_1$  = LSE of  $\beta_1$  and is unbiased
- MLE of  $\sigma^2$  is less than the unbiased estimator of  $\sigma^2$ , but is asymptotically unbiased

## Distribution of $\hat{\beta}_1$

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, 2, \dots, n,$$

- Assumptions

- $x_i$ 's are known constants,
- $\epsilon_i \sim N(0, \sigma^2)$  independently for  $i = 1, 2, \dots, n$

- Therefore,  $y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$



$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \sum_{i=1}^n c_i y_i$$

where  $c_i = \frac{(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$ , and then  $\hat{\beta}_1$  follows a normal distribution.

- $\hat{\beta}_1 \sim N(\beta_1, \sigma^2 / S_{xx})$ .



## Testing (Two-sided test of $\beta_1$ )

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, 2, \dots, n,$$



$H_0 : \beta_1 = 0$  (no linear relationship) v.s.

$H_1 : \beta_1 \neq 0$  (linear relationship does exist between  $x$  and  $y$ )

■ Test statistic:

$$t = \frac{\hat{\beta}_1 - \beta_1}{S/S_{xx}^{1/2}} \sim t_{n-2} \text{ if } H_0 \text{ is true}$$

Remark 2.3

■ Decision rule: reject  $H_0$  if  $|t| > t_{\alpha/2, n-2}$ .

Remark 2.3

$H_0: \beta_1 = 0 \quad H_1: \beta_1 \neq 0$

Use the fact

$$\hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{S_{xx}})$$

OR

$$\frac{\hat{\beta}_1 - \beta_1}{(\sigma^2 / S_{xx})^{1/2}} \sim N(0, 1)$$

$\sigma^2$  - unknown  $\sigma^2$  is replaced by  $\hat{\sigma}^2 = s^2$

$$\frac{(n-2)s^2}{\sigma^2} \sim \chi^2_{n-2}$$

} the proof will be given later

$\hat{\beta}_1$  and  $s^2$  are independent

$$t = \frac{\hat{\beta}_1 - \beta_1}{(\sigma^2 / S_{xx})^{1/2}} \left/ \left[ \frac{(n-2)s^2}{\sigma^2} / (n-2) \right]^{1/2} \right. \sim t_{n-2}$$

$$\frac{N(0,1)}{(\chi^2_{n-2} / (n-2))^{1/2}}$$

$$= \frac{\hat{\beta}_1 - \beta_1}{s / S_{xx}^{1/2}} \underset{H_0}{\sim} t_{n-2}$$

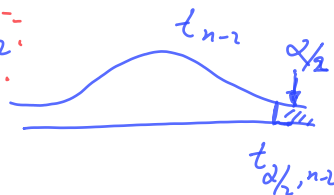
p-value

$$= 2P_r(t_{n-2} \geq |t|)$$

C.I. of  $\beta_1$  with  $(1-\alpha)$  level:

$$\hat{\beta}_1 \pm s / S_{xx}^{1/2} \cdot t_{\alpha/2, n-2}$$

reject  $H_0$  if  $|t| \geq t_{\alpha/2, n-2}$



## Two-sided test and confidence interval of $\beta_1$

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, 2, \dots, n,$$

■

$$H_0 : \beta_1 = k \quad \text{v.s.} \quad H_1 : \beta_1 \neq k \quad (k \text{ is a constant})$$

■ What are the test statistic and decision rule?

■ What are the confidence interval of  $\beta_1$ ?

## Distribution of $\hat{\beta}_0$

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2) \quad i = 1, 2, \dots, n,$$

- $\hat{\beta}_1 \sim N(\beta_1, \sigma^2 / S_{xx})$ .
- $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$  also follows a normal distribution

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2 \left[ \frac{1}{n} + \frac{\bar{x}^2}{\sum_i (x_i - \bar{x})^2} \right]\right)$$

## Estimation of the mean of the response variable for a given level of $x$

### ■ Example

- $y$  (in \$000) – house price,  $x$  (square feet) – house size
  - Estimate the average house price for houses with 2000 square feet.
- Let  $x_h$  be the level of  $x$  for which we wish to estimate the mean response, then

$$y_h = \beta_0 + \beta_1 x_h + \epsilon_h,$$

the mean response is  $E(y_h) = \beta_0 + \beta_1 x_h$ .

- The estimation of  $E(y_h)$  is  $\hat{y}_h = \hat{\beta}_0 + \hat{\beta}_1 x_h$ , with distribution

$$\hat{y}_h \sim N \left( \beta_0 + \beta_1 x_h, \sigma^2 \left[ \frac{1}{n} + \frac{(x_h - \bar{x})^2}{\sum_i (x_i - \bar{x})^2} \right] \right)$$

Remark 2.4 model  $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, i=1, \dots, n$   
 $\varepsilon_i \sim N(0, \sigma^2)$   
 $\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2$

At a new data point  $x_h$  ( $= 2000$ )

\* The average house price at  $x = x_h$ .

$$E \hat{y}_h = \beta_0 + \beta_1 x_h$$

$$\hat{y}_h = \hat{\beta}_0 + \hat{\beta}_1 x_h$$

c.i. of  $E \hat{y}_h$  or  $\beta_0 + \beta_1 x_h$  is of interest!

$$\hat{y}_h \sim N(, )$$

$$E \hat{y}_h = \beta_0 + \beta_1 x_h$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\text{Var}(\hat{y}_h) = \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_h)$$

$$= \text{Var}(\bar{y} + \hat{\beta}_1 (x_h - \bar{x}))$$

$$= \text{Var}(\bar{y}) + (x_h - \bar{x})^2 \text{Var}(\hat{\beta}_1)$$

$$+ 2(x_h - \bar{x}) \text{cov}(\bar{y}, \hat{\beta}_1)$$

$$= \frac{\sigma^2}{n} + \frac{(x_h - \bar{x})^2}{S_{xx}} \sigma^2 = \left( \frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_{xx}} \right) \sigma^2$$

$$\hat{y}_h \sim N\left(\beta_0 + \beta_1 x_h, \left(\frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_{xx}}\right) \sigma^2\right)$$

It is easy to construct the c.i. of

$$E(\hat{y}_h) = \beta_0 + \beta_1 x_h = E(y_h)$$

## Confidence interval for $E(y_h)$

$$E(y_h) - \hat{y}_h \sim N \left( 0, \sigma^2 \left[ \frac{1}{n} + \frac{(x_h - \bar{x})^2}{\sum_i (x_i - \bar{x})^2} \right] \right)$$

Two-sided  $100(1 - \alpha)\%$  C.I. for  $E(y_h)$  is

$$\left( \hat{y}_h - t_{\alpha/2, n-2} S \sqrt{\frac{1}{n} + \frac{(x_h - \bar{x})^2}{\sum_i (x_i - \bar{x})^2}}, \hat{y}_h + t_{\alpha/2, n-2} S \sqrt{\frac{1}{n} + \frac{(x_h - \bar{x})^2}{\sum_i (x_i - \bar{x})^2}} \right)$$

Remark 2.5 Prediction of a new observation.

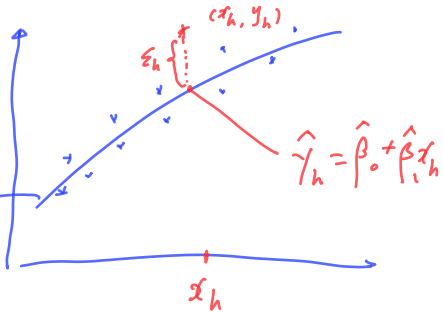
c.i. of  $E \hat{y}_h = \beta_0 + \beta_1 x_h$

$$y_h = \beta_0 + \beta_1 x_h + \varepsilon_h$$

$$y_h - \hat{y}_h = \beta_0 + \beta_1 x_h + \varepsilon_h - \hat{y}_h$$

$$= \underbrace{(E y_h - \hat{y}_h)}_{\text{independent}} + \varepsilon_h$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$



$$\sim N(0, \text{Var}(E y_h - \hat{y}_h) + \sigma^2)$$

$$\sim N(0, \left( \frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_{xx}} \right) \sigma^2 + \sigma^2)$$

c.i. of  $y_h \Rightarrow$  Predictive interval of  $y_h$

\* Prediction of  $y_h$  and  $E y_h$  are the same

$$\hat{y}_h = E \hat{y}_h = \hat{\beta}_0 + \hat{\beta}_1 x_h$$

\* but the variances of

$$y_h - \hat{y}_h \quad \text{and} \quad E(y_h) - \hat{y}_h \quad \text{are different}$$

$$\text{Var}(y_h - \hat{y}_h) = \text{Var}(E(y_h) - \hat{y}_h) + \underline{\text{Var}(\varepsilon_h)}$$

\* Predictive interval of  $y_h$  is wider than the c.i. of  $E(y_h)$ .

✗



## Prediction of a new observation $y_h$

### ■ Example

- $y$  (in \$000) – house price,  $x$  (square feet) – house size
- Estimate the house price for **an individual** house with 2000 square feet.
- It means we wish to estimate the response  $y_h$  given  $x_h$

$$y_h = \beta_0 + \beta_1 x_h + \epsilon_h,$$

- The estimation of  $y_h$  is still  $\hat{y}_h = \hat{\beta}_0 + \hat{\beta}_1 x_h$ , but

$$y_h - \hat{y}_h \sim N \left( 0, \sigma^2 \left[ 1 + \frac{1}{n} + \frac{(x_h - \bar{x})^2}{\sum_i (x_i - \bar{x})^2} \right] \right)$$

## Confidence interval for a new observation $y_h$

Predictive interval

Two-sided  $100(1 - \alpha)\%$  C.I. for  $y_h$  is

$$\left( \hat{y}_h - t_{\alpha/2, n-2} S \sqrt{\underset{\text{red}}{1} + \frac{1}{n} + \frac{(x_h - \bar{x})^2}{\sum_i (x_i - \bar{x})^2}}, \right. \\ \left. \hat{y}_h + t_{\alpha/2, n-2} S \sqrt{\underset{\text{red}}{1} + \frac{1}{n} + \frac{(x_h - \bar{x})^2}{\sum_i (x_i - \bar{x})^2}} \right)$$

We also call it as a **predictive interval**.

## Analysis of variance approach to regression analysis

- Partitioning of Total Sum of Squares (SST)

$$\begin{aligned} SST &= \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 \\ &= SSE + SSR \end{aligned}$$

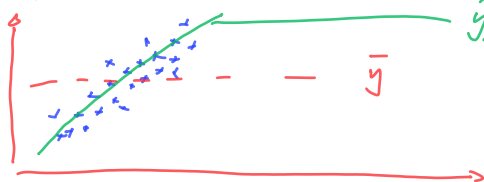
where SSE=sum of squares of residual, SSR=sum of squares due to regression.

- OR

Total Variation = Unexplained Variation + Explained Variation

Remark 2.6

$$D = \{ (x_i, y_i), i=1, \dots, n \}$$



$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Total variation of  $y_i$  (SST)

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$= \sum_{i=1}^n (y_i - \hat{y}_i + \hat{y}_i - \bar{y})^2$$

$$= \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + 2 \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y})$$

$$= \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$= SSE + SSR$$

sum of squares  
of residuals

sum of squares  
due to regression

unexplained  
variation

Explained variation  
by the model

$$R^2 = \frac{SSR}{SST} \quad \text{— \% of the variation can be explained by the model!}$$

R-squared. coefficient of determination

$$\hat{y}_i - \bar{y} = \hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y}$$

$$= \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_i - \bar{y}$$

$$= \hat{\beta}_1 (x_i - \bar{x})$$

$$\sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y})$$

$$= \sum_{i=1}^n (y_i - \hat{y}_i) \hat{\beta}_1 (x_i - \bar{x})$$

$$= \hat{\beta}_1 \sum_{i=1}^n (y_i - \hat{y}_i) x_i$$

$$= \hat{\beta}_1 \bar{x} \sum_{i=1}^n (y_i - \hat{y}_i)$$

ANOVA table in regression model

two equations to obtain the estimation of  $\hat{\beta}_0$  and  $\hat{\beta}_1$

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (*)$$

$H_0: \beta_1 = 0$ : meaning the model with no covariate fits the data as well as the model (\*)

$H_1: \beta_1 \neq 0$ : meaning the model (x) (with the covariate) fits the data better than the intercept only model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$$

$$H_0: \beta_1 = \beta_2 = 0$$

$$H_1: \text{at least one of } (\beta_1, \beta_2) \neq 0$$

|            | sum of squares (S.S.) | d.f.    | M.S.              | F       |
|------------|-----------------------|---------|-------------------|---------|
| Regression | SSR                   | p       | MSR = SSR/p       | MSR/MSE |
| Error      | SSE                   | n-(p+1) | MSE = SSE/(n-p-1) |         |
| Total      | SST                   | n-1     |                   |         |

under  $H_0: \beta_1 = 0$ .

$$\frac{SSR/p}{SSE/(n-p-1)} = \frac{MSR}{MSE} \stackrel{H_0}{\sim} F_{p, n-p-1}$$

\* SSR and SSE are independent

$$* SSE/\sigma^2 \sim \chi^2_{n-p-1}$$

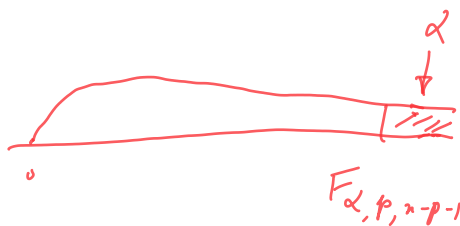
$$* SSR/\sigma^2 \sim \chi^2_p$$

$$\text{Reject } H_0 \text{ if } F = \frac{MSR}{MSE} > F_{\alpha, p, n-p-1}$$

$$\text{OR } P\text{-value} < 0.05$$

P-value

$$= P_r(F_{p, n-p-1} \geq F)$$



## Analysis of variance (ANOVA) table

|            | Sum of Squares (SS) | Degrees of freedom (df) | Mean squares (MS)         | $F$               |
|------------|---------------------|-------------------------|---------------------------|-------------------|
| Regression | SSR                 | 1                       | $MSR = \frac{SSR}{1}$     | $\frac{MSR}{MSE}$ |
| Error      | SSE                 | n-2                     | $MSE = \frac{SSE}{(n-2)}$ |                   |
| Total      | SST                 | n-1                     |                           |                   |

- Test  $H_0 : \beta_1 = 0$  (**no linear relationship**) v.s.  $H_1 : \beta_1 \neq 0$  (**linear relationship does exist** between  $y$  and  $x$ )
- Test statistics  $F = \frac{MSR}{MSE} \sim_{H_0} F_{1,n-2}$
- Reject  $H_0$  if  $F > F_{\alpha,1,n-2}$ .

## The coefficient of determination

- The coefficient of determination OR R-squared is defined

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

- The proportion of the variation can be explained by the model:  $0 \leq R^2 \leq 1$ .
- Coefficient of correlation (true for simple linear regression only)

$$r = \pm\sqrt{R^2}$$